3.2 Types of mitigation measures

Potential mitigation measures that could be implemented in the framework of this analysis are numerous (Turner and Schuster, 1996). Here we follow the classification proposed by the SafeLand project (https://www.ngi.no/eng/Projects/SafeLand); the SafeLand report (SafeLand, 2012) is attached as a supportive document. Table 12 shows that the Total Risk associated with landslides can be mitigated by reducing:

- the Hazard – H- (i.e. the probability of occurrence of one or more phenomena);
- the Vulnerability – V - (i.e. the degree of loss to the elements at risk for a given hazard);
- the Elements at risk – E- (i.e. their number and/or specific value).

Table 12 makes the difference between structural and non-structural measures. In general terms, it means:

- "structural" measures include, but are not limited to drainage, erosion protection, channelling, vegetation, ground improvement, barriers such as earth ramparts, walls, artificial elevated land, anchoring systems and retaining structures; buildings designed and/or placed in locations to withstand the impact forces of landslides and to provide safe dwellings for people, and escape routes;
- "non-structural" or more generally "consequence reducing measures" include, but are not limited to: retreat from hazard, land-use planning, early warning, public preparedness, (escape routes, etc.) and emergency management.

Table 12: **General classification of mitigation measures (SafeLand, 2012).**

Classification		Component of risk addressed	Brief description	Notes and other terms used
STRUCTURAL ↑ NON STRUCTURAL ↓	Stabilization	Hazard (H)	engineering works to reduce the probability of occurrence of landsliding	Preventive, remedial, hard, soft, active stabilization
	Control	Vulnerability (V)	engineering works to protect, reinforce, isolate the elements at risk from the influence of landsliding	Preventive, hard, soft, passive stabilization
	Avoidance	Elements (E)	temporary and/or permanent reduction of exposure through: warning systems and emergency evacuation or safe sheltering, land-use planning and/or relocation of existing facilities	Direct temporary and/or permanent reduction of the number and/or value of elements at risk. Monitoring and warning or alarm systems and associated civil protection procedures, often described as reducing vulnerability, in actual fact operate through temporary, selective avoidance.
	Tolerance	Elements (E)	Awareness, acceptance and/or sharing of risk	Indirect reduction of the number and/or value of elements at risk